

A Market-Neutral Multi-Factor Equity Strategy

Momentum, profitability and investment premia in the 500 largest US stocks

At a glance: out of sample, January 1995 – December 2025, net of 10 bp trading costs

- Dollar-neutral long-short equity book, rebalanced monthly, built with a Barra-style risk model and Gurobi
- **Return/vol 1.43** against 0.66 for the market (Sharpe ratio **0.76** against 0.53)
- Maximum drawdown -6% against -55% Correlation with the market $+0.04$
CAPM alpha t -stat 4.2

Data, universe and backtest protocol

- **Data:** CRSP monthly and daily returns (`msf_v2`, `dsf_v2`), Compustat annual fundamentals, point-in-time GICS sectors, market and risk-free rate from `MktRf.csv` (Ken French)
- **Universe:** each month, the 500 largest US common stocks with price $\geq \$5$ and at least 24 months of history; incumbents stay while their size rank is $\leq 550 \Rightarrow$ about 80% of US market cap, 5.5 names replaced per month
- **Walk-forward backtest:** at each month-end d , estimate everything with data up to d (60-month windows), form the portfolio, hold it over month $d + 1 \Rightarrow 372$ out-of-sample months
- **No look-ahead:** Compustat data used only 6 months after the fiscal year-end
- **Costs and measurement:** 10 bp per unit of turnover; performance measured on daily returns of the monthly-held books (ratios $\times \sqrt{252}$)

From stock characteristics to an alpha

Signals (higher = expected to outperform)

mom	12–1 month return
gp	(revenue – COGS) / total assets
ag	– annual growth of total assets
rev	– last month's return (<i>tested, dropped</i>)

Score: winsorize at ± 3 std, z-score, demean within each GICS sector $\Rightarrow z_{i,d}$

Grinold's rule

$$\alpha_{i,d} = \text{IC} \times \sigma_{i,d} \times z_{i,d}$$

IC = 0.02 (prior); $\sigma_{i,d}$ = the stock's specific volatility from the risk model.

The strategy: one optimized book per signal (mom, gp, ag), held in equal parts.

Risk model: Barra-style factor model, re-estimated every month

Cross-sectional regression (weights $\sqrt{\text{cap}}$) of next month's returns on today's exposures:

$$r_{i,d+1} = \sum_{s=1}^{10} \mathbf{1}\{i \in s\} f_{s,d+1} + \sum_k x_{ik,d} f_{k,d+1} + u_{i,d+1}$$

- **17 factors**: 10 GICS sectors; styles beta (252 daily returns), size (log cap), residual volatility; and **the four signals**, so the model sees that their tilts are systematic
- Exposures winsorized, cap-weighted mean 0 and unit std (Barra convention)
- Covariance at each date d , from data up to d :

$$V_d = X_d F_d X_d^\top + D_d$$

F_d : sample covariance of the last 60 monthly factor returns D_d : diagonal, each stock's residual variance over 60 months

Portfolio construction (Gurobi, every month)

$$\begin{aligned} \max_w \quad & \alpha_d^\top w - \kappa \sum_i |w_i - w_i^{\text{prev}}| \\ \text{s.t.} \quad & y = X_d^\top w \\ & y^\top F_d y + \sum_i D_i w_i^2 \leq \sigma_{\text{target}}^2 / 12 \\ & \mathbf{1}^\top w = 0 \\ & |y_s| \leq 2\% \text{ for each sector } s \\ & |y_k| \leq 0.10 \text{ for } k \in \{\text{beta, size, res. vol}\} \\ & |w_i| \leq 2\%, \quad \sum_i |w_i| \leq 4 \end{aligned}$$

$\sigma_{\text{target}} = 6\%$ a year; $\kappa = 10$ bp (the trading cost).

Factor exposures: neutral to the market, sectors, size and volatility. Signal exposures are **free**, but their risk counts in the budget.

Regularization

- factor structure: 17 factors instead of a 500×500 sample covariance
- position, leverage and exposure bounds
- L_1 turnover penalty
- winsorized inputs, fixed IC prior
- explicit shrinkage of F and D : tested, did not help

The signals are systematic: price their risk, don't hedge it

Stocks favored by a signal move together: a signal tilt is one factor bet, not 500 independent bets.

Signal	Risk forecast bias	Realized / target vol	Net Sharpe
	base → aware model	blind → aware	blind / aware free / aware neutral
mom	2.14 → 1.13	3.01 → 1.14	0.22 / 0.31 / 0.04
gp	1.57 → 0.93	1.82 → 1.06	0.31 / 0.46 / 0.04
ag	1.25 → 0.92	1.60 → 1.01	0.15 / 0.18 / -0.03

Bias = std of realized return / predicted vol of the signal tilt (1 = accurate). Blind: risk model without the signals. Aware free: signals in the risk model, exposure free. Aware neutral: signal exposure held near 0. No turnover penalty in this table.

Takeaway: with the signals as risk factors, realized risk matches the target and the Sharpe is highest; hedging them out removes the premium.

Transaction costs: a turnover penalty in the objective

Net Sharpe ratio (monthly turnover in brackets), aware model, 10 bp per unit traded:

Signal	$\kappa = 0$	$\kappa = \text{cost}$	$\kappa = 3 \times \text{cost}$
mom	0.31 (1.58)	0.42 (0.27)	0.27 (0.12)
gp	0.46 (0.70)	0.48 (0.11)	0.32 (0.06)
ag	0.18 (0.74)	0.44 (0.18)	0.64 (0.10)
rev	-0.60 (3.71)	-0.02 (1.09)	-0.11 (0.45)

Takeaway: charging the true cost ($\kappa = \text{cost}$) cuts turnover 3–6 $\times$ and raises net Sharpe. Short-term reversal only works before costs, so it is dropped. All final books use $\kappa = \text{cost}$.

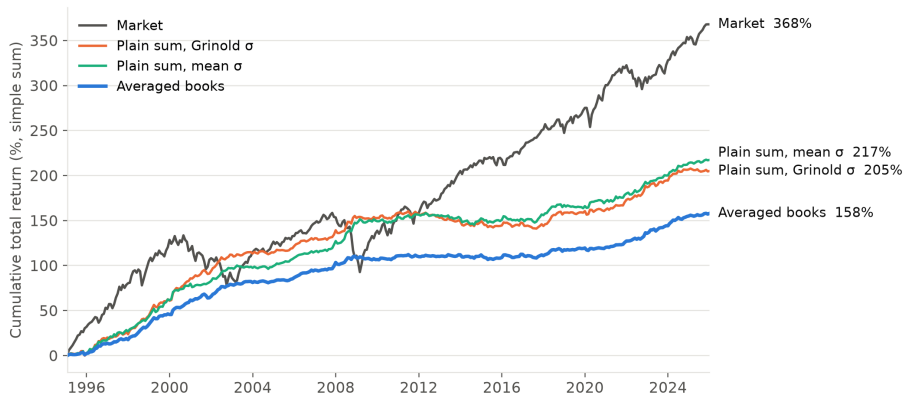
Performance, January 1995 – December 2025 (net of costs)

Book	Return / vol	Sharpe	Ann. return	Ann. vol	Max drawdown
Market (benchmark)	0.66	0.53	12.5%	19.0%	−55%
Averaged books (mom, gp, ag)	1.43	0.76	5.1%	3.6%	−6%
Plain sum of Grinold alphas	1.12	0.72	6.6%	5.9%	−19%
Plain sum, cross-sectional mean σ	1.20	0.80	7.0%	5.9%	−14%

- Almost no market exposure: correlation +0.04, beta ≈ 0 , CAPM alpha t -stat 4.2
- The three books diversify each other, so the averaged book runs at 3.6% vol; scaled to 6% vol it earns 4.6% a year over cash
- Plain-sum variants: one optimization on $\alpha_{\text{mom}} + \alpha_{\text{gp}} + \alpha_{\text{ag}}$, using each stock's σ_i (Grinold) or the cross-sectional mean $\bar{\sigma}$

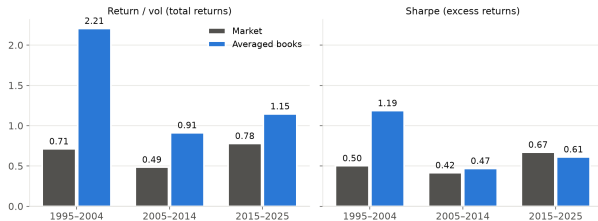
Daily returns, ratios $\times \sqrt{252}$. Return/vol uses total returns (risk-free rate not subtracted); the Sharpe ratio subtracts it (2.4% a year on average).

Cumulative performance



Cumulative sum of monthly total returns, net of costs. The market carries about 5 times the averaged book's volatility.

Across decades, and takeaways



- Return/vol above the market in every decade; Sharpe above it in 1995–2004 and 2005–2014, slightly below in 2015–2025 (0.61 vs 0.67)
- Near-zero beta and correlation: a diversifier for an equity portfolio
- Returns come from the momentum, profitability and investment premia, with market, sector, size and volatility risk hedged
- Risks: factor crashes (momentum in 2009) and weaker premia in large caps since 2005